BarrierNet: Learning Safe Robot Control with Differentiable CBF

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Background

- Learned policies perform well but can fail in unseen scenes.
- In safety-critical robotics, unsafe actions are unacceptable.
- Goal: combine learning flexibility with formal safety guarantees.
- BarrierNet treats safety as a trainable module rather than a fixed rule.

Related Work and Gap

- Model-based safe control is often expensive.
- Classical CBF filters are safe but hand-tuned and conservative.
- Differentiable layers exist, but most do not adapt the safety rule itself.

Research Gap

Need: an adaptive safety filter that remains formally safe.

Research Aim

- Plug into a nominal neural controller.
- Correct unsafe actions with a differentiable CBF-QP.
- Learn safety adaptation from data.
- Keep the final action inside a certified safe set.

Main Takeaway

Learn the controller *and* the safety adaptation rule together, instead of relying on one fixed barrier design for every situation.

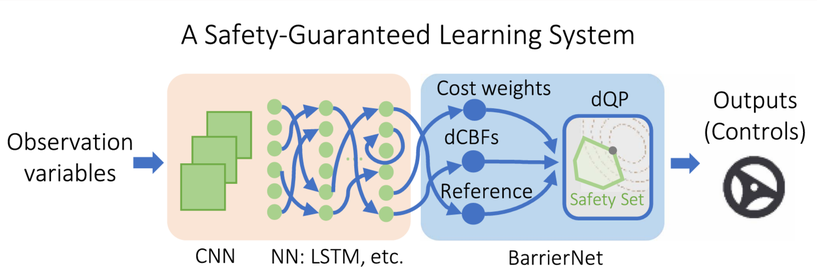
Keywords

Safe learning, CBF, Differentiable QP, Robot control

Poster Summary

- **Problem:** fixed safety filters are often too conservative.
- **Method:** learn adaptive CBF parameters through a differentiable QP layer.
- **Outcome:** safer control with better task performance.

BarrierNet Method



Pipeline:

1. Nominal policy proposes an action.
2. BarrierNet predicts adaptive CBF parameters.
3. A differentiable QP projects to the nearest safe action.

Core Idea

For a control-affine system

$$\dot{x} = f(x) + g(x)u,$$

BarrierNet enforces a CBF condition of the form

$$L_f h(x) + L_g h(x) u + \alpha(h(x)) \geq 0,$$

and computes the safe action by solving

$$u^\star = \arg\min_u \|u - u_{\text{nom}}\|^2$$

subject to the safety and actuator constraints.

Key idea: learn the barrier adaptation, not only the policy.

Why This Matters

- Fewer unnecessary interventions.
- Better performance than fixed barriers.
- Joint optimization of policy and safety.

Training Perspective

- The optimization layer is differentiable.
- Training encourages policies that are easier to keep safe.

Compared with Fixed CBF Filters

- Using one setting for all states.
- Adapts the safety margin online.
- Result: safety-performance balance.

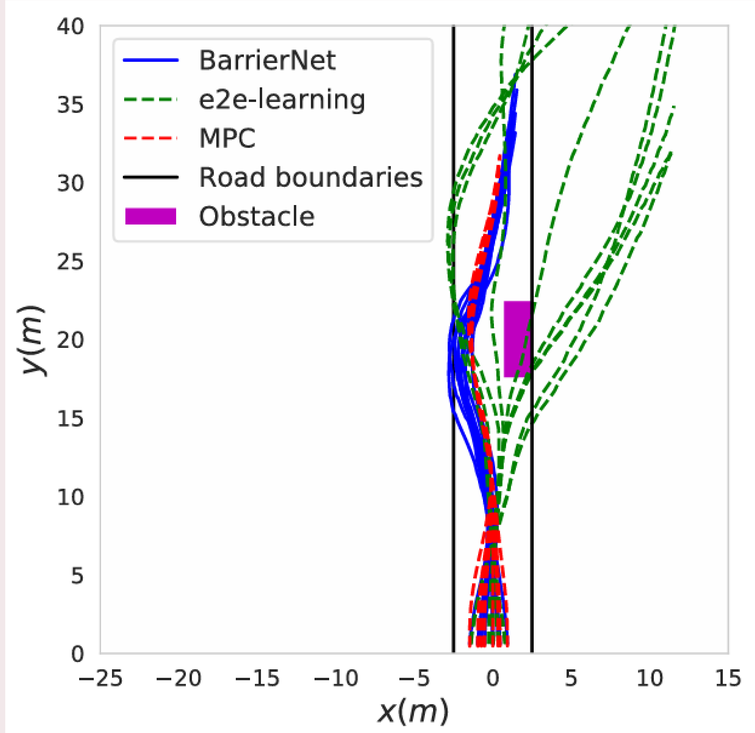
Take-Home Insight

BarrierNet does not replace CBF theory. It makes the *barrier design* adaptive enough for learning-based control.

Code

https://github.com/Weixy21/BarrierNet

Experimental Results



Observed outcome from the paper:

- Safer than unconstrained learned control.
- Less conservative than fixed CBF baselines.
- More robust under changing environments and input limits.

Contributions

1. Differentiable CBF safety layer.
2. Adaptive barrier learning.
3. Better safety-performance tradeoff.

Limitations and Discussion

- Guarantees still depend on model quality.
- QP feasibility can remain difficult under tight limits.

Practical Significance

- Relevant to driving, mobile robots, and manipulators.
- Can wrap an existing learned controller.

Reference

Xiao, W., Cassandras, C. G., and Belta, C. *BarrierNet: Learning Safe Robot Control with Differentiable Control Barrier Functions.*

Poster Code

https://github.com/edyou25/PolyU_Poster